

How to Build Trust Into 5G

Fall 2021 Final Report

The University of Texas at Dallas

Team 3

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Project Summary

Description

There is a growing need to move 5G to a decentralized, autonomous network due to 5G's growing size, multi-layer, and multi-party nature, and usage of 5G slicing for service optimization. This decentralization involves sharing parts of 5G's infrastructure with third parties, which requires an effective security system to verify the trust of the third parties to ensure that only trustworthy data and data sources are in the 5G network. Remote attestation, which verifies a device's integrity by prompting the device to send its identifying information to a verifying agent, has been proposed as one of several possible ways to build trust in 5G. The challenge is figuring out a proper way to successfully implement mutual remote attestation into the 5G network. One of the facets of this project will be applying research findings to roaming and handover test cases.

Motivation

Telecommunication is a core part of our society that has radically changed the way of life. 5G, the next generation of wireless technology, has considerable improvements over its predecessor both in terms of speed and reliability. This has opened up many possibilities which we could've only dreamt of. As such contributing to the development of this technology might not only have an immediate effect on 5G but also on its future generations.

Team Responsibilities

- Understand the architecture of 5G networks and brush up on tools required like WireShark and Git
- Understand the current state of attestation protocols and how they can be applied to a decentralized system
- Design a novel solution for secure attestation and communication in a decentralized 5G network
- Meet with the sponsor weekly for reports and progress updates

Broader Impact

As telecommunication networks are moving to a more decentralized architecture the need for mutual attestation that can be used on decentralized networks is becoming more important. This research will help lead to a better system for devices on these networks to create trust. This applies not only to 5G but can extend and be applied to any decentralized network and future generations.

Organization

Deliverables

- Proposal
- Midterm Report and Presentation
- Final Report and Presentation
- Documentation of research

Plan

The team will meet with the sponsor every Monday and Thursday from 12:00 PM - 1:00 PM to discuss the progress of the research project. The team may schedule additional meetings if necessary.

Schedule

September 10	- Proposal and Literature Review Due
September	- Learning current 5G and mutual attestation
October	- Setup testing environment
October 21	- Midterm Report and Presentation Due
November	- Design workflow for existing attestation protocol to implement into 5G
December 9	- Final Report and Presentation Due

Total Budget

This Project has no monetary cost. The group intends to work around nine to eleven hours per week on this project along with weekly meeting updates.

Team Approach and Justification

The team plans to familiarize themselves with 5G and modern attestation protocols by researching both topics and getting hands-on experience with 5G. The team will then explore and analyze different attestation workflows and protocols to use in the 5G network. Hands-on development will then be used to perform tests on implementing these workflows and protocols into the 5G network for the roaming/handover use case.

We plan on using Open5GS to simulate and test the 5G network. We will use Wireshark to collect information data during tests to see actual test results. The final product will be an attestation protocol that works smoothly and securely in the decentralized 5G network architecture, guaranteeing accurate verification of trust.

The results of this approach will help determine the type of remote attestation workflows and protocols that could work in building trust in 5G for roaming and handover scenarios. This in turn will improve the overall understanding of how remote attestation can be effectively implemented in 5G in general, as well as how feasible the implementation would be.

Midterm Progress Update

Open5GS + UERANSIM + Docker Testbed

Our team set up a testing environment using Open5GS and UERANSIM to simulate communications over a 5G network. We chose Open5GS as the 5G core framework because it is open-source and it has more documentation and active maintenance compared to other frameworks. Open5GS also supports various types of 5G network deployment, including NSA (Non-Standalone / Hybrid) and SA (Standalone) modes.

Although Open5GS provides the core 5G network, they do not provide implementations for either the base stations or the user equipment, both of which are required to test the roaming-handover use case. We chose to use the UERANSIM project because it provides software-based implementations for both the gNB (g-Node-B) and UE (User-Equipment). To capture the communication between the different components of the 5G Network, we integrated Wireshark into the testbed.

As 5G is moving towards a more “cloud”-based, we wanted our testbed to have a similar test environment. We could deploy these services in two ways: Kubernetes or Docker-Compose. We initially chose Kubernetes as it is the best container orchestration technology available right now, but setting up the cluster and making sure everything worked was very time-consuming.

We needed a much more straightforward and manageable way to deploy our testbed, so we chose the Docker route. We have set up our environment such that all the core 5G components, gNBs, and the UE share the same Docker network and that Wireshark can be easily used to analyze traffic on this network.

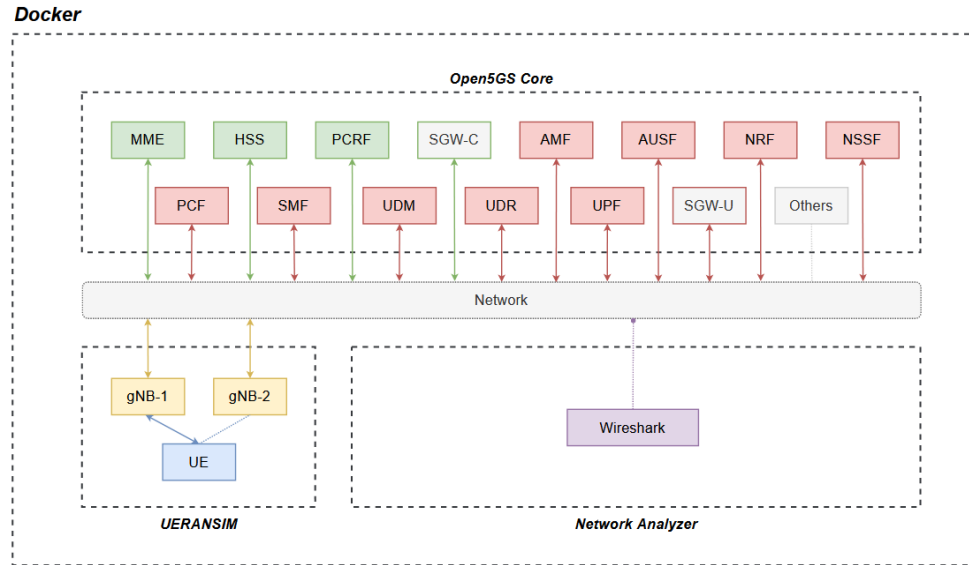


Figure 1: Testbed Deployment Diagram

Modeling Attestation in the Handover Case

Our team created Figures 2 through 4 to illustrate how remote attestation would be implemented in the handover test case.

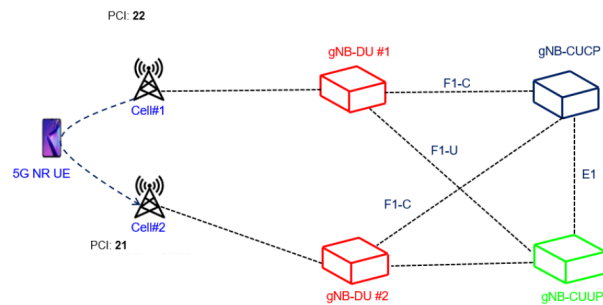


Figure 2: 5G Inter-gNB-DU Handover

Figure 2 depicts the handover case where the UE connected to source cell 1 with gNB-DU 1 moves to cell 2 with gNB-DU 2. The interfaces among the gNB components make use of protocols used for signaling service with F1AP protocol for F1 interface and E1AP protocol for E1 interface. For this type of handover, no signaling is exchanged with the 5G core as both gNB-DU lie under the same gNB-CU. Here the gNB is split into a Distributed Unit (DU) and Control Unit (CU), which further splits into Control Plane CU-CP and User Plane CU-UP to adapt itself for different RAN performance requirements and location scenarios, as well as to have a distinguished user and control plane traffic management.

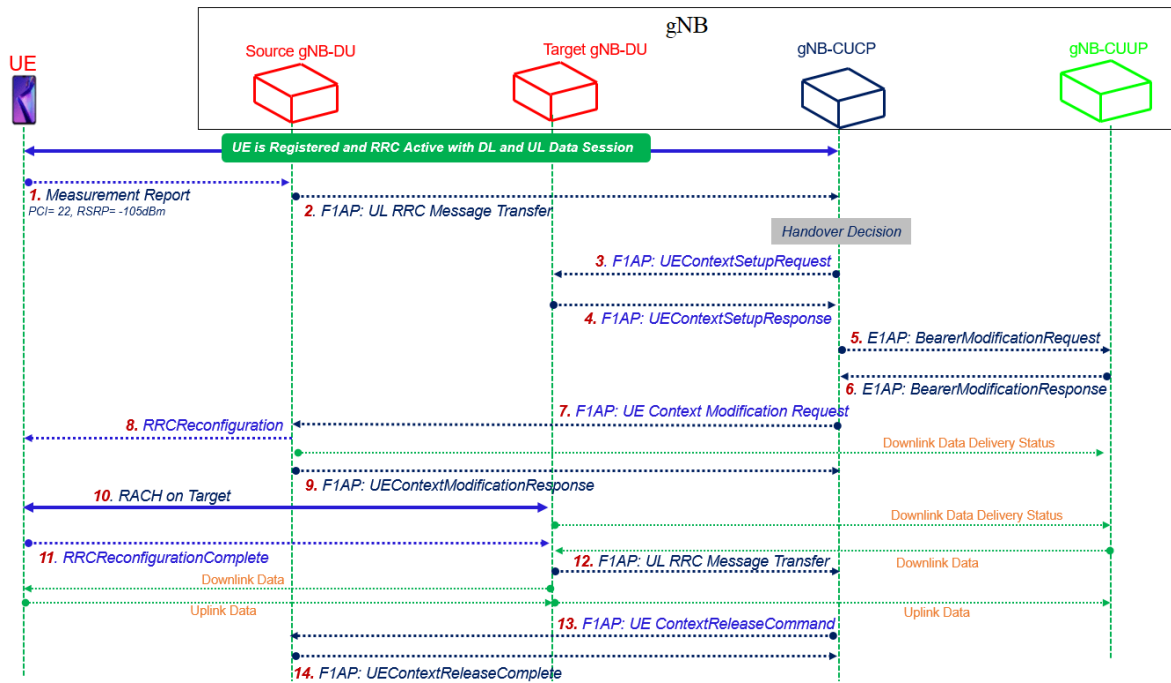


Figure 3: 5G Inter-gNB-DU Handover Call flow

Figure 3 shows the handover call flow for inter-gNB-DU mobility. In 5G, handover decisions are based on the measurement report provided by UE. The measurement report contains the signal strength of both serving and neighboring cells. Along with the measurement report and the serving network name (SNN), the source gNB-DU sends a UL RRC Message Transfer to gNB-CU-CP to make a handover decision. After deciding to go with the handover, gNB-CU-CP sends a UE Context setup message to target gNB-DU to make a UE Context and set up SRBs and DRB's bearers. The gNB-CU-CP sends a bearer modification request to gNB-CU-UP, indicating it modifies a list of DRB's. Then gNB-CU-CP sends a message to source gNB-DU to stop data transmission with the UE. The source gNB-DU then conveys the handover decision to UE by sending an RRC configuration message. With this message received by the UE, it initiates the RACH procedure. The RACH procedure tries to acquire the uplink synchronization and cell ID of the target gNB-DU and sends a reply message containing successful completion of RRC

Reconfiguration. The message is conveyed to gNB-CU-CP via gNB-DU, allowing gNB-CU-CP to send a UE context release message to source gNB-DU to release a UE Context with the proper value. This step marks the completion of the handover.

Choices of Attestation Protocols to Test

Collective Remote Attestation (CRA) Protocol

This attestation protocol uses a unique measurement that is obtained from an entire network of IoT devices. This measurement is mainly used to describe the collective status of the entire network, but it can also provide information on the status of each device in the network. The main advantage of the CRA protocol is it can be scaled easily for larger networks, unlike other remote attestation protocols. However, we will need to distinguish attestation scenarios caused by intermittent connections from scenarios caused by IoT devices being compromised. It is also important that we choose a key mechanism that will result in a good balance of efficiency and security [6].

Authentication and Key Agreement (AKA) Protocol

The AKA protocol is standardized for the 5G network and it is one of the primary authentication methods defined for 5G. These protocols involve the subscribers, the serving networks (SNs) with base stations in subscribers' vicinity, and Home Networks (HNs) corresponding to the subscribers' carriers. The goal of the AKA protocol is to allow the subscribers and the HNs to authenticate each other mutually. Another goal of AKA is to let the subscribers and the SNs establish a session key. This protocol aims to authenticate subscribers and their carriers mutually and establish a secure channel to protect subsequent communication.

Out of the two protocols, we decided to use the AKA protocol.

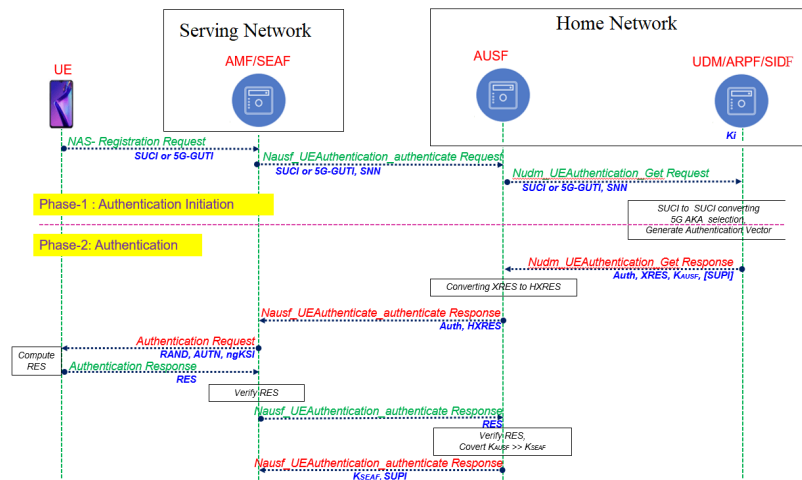


Figure 4: AKA Protocol in Handover Case

Figure 4 shows the AKA attestation flow among the UE, Serving Network, and Home Network. In Phase 1, the authentication procedure is initiated when UE performs a request, such as a registration request. For such a request, the UE needs to send Unique identifiers to its serving network to identify itself using a Subscription Identifier or a Global Unique Temporary Identifier. Unlike 4G/LTE, these identifiers are sent in a concealed format to protect the identity of UE. The serving network includes the Access and Mobility Management Function (AMF), which is responsible for handling connections and mobility tasks. The Security Anchor Function (SEAF), which resides within AMF, acts as an intermediary during the authentication process between the UE and the Home Network. It then forwards the received UE identifier along with the serving network name to AUSF. The AUSF or authentication Server Function is responsible for deciding on authenticating the UE, which requires security credentials such as key and authentication data, which are provided by Unified Data Management (UDM). The ARPF selects the authentication procedure based on UE identity and policy. SIDF is used to decrypt the UE concealed identity. The UDM then sends the challenge, keys, and response to AUSF, which stores the keys and hashed value of response and sends the challenge to UE via SEAF. Meanwhile, the SEAF also stores the hashed response. Upon receiving the challenge, the UE generates the response. The SEAF verifies this response. However, if the response does not match the hashed response, then the SEAF can reject the authentication. Furthermore, the response is sent to the AUSF to have more control over this process. Only when the AUSF performs successful verification does it provide the SEAF with the keys for secure communication with UE.

Final Progress Update

Change of Project Goals

As the project progressed, the team concluded that developing and describing the complete protocol and implementing the code to test and simulate the protocol would take longer than planned. Due to this conclusion, we changed the final deliverable to be a fully described protocol with diagrams for message call flow and message contents and forgo the implementation for the next team to work on this project. We include in our final deliverables our initial work on setting up a testbed and the different tools and setup commands that had to be used to get it running. This should help jump-start the next team into modifying these tools to implement the protocol into the testbed. Our change of plans was caused due to the team's unfamiliarity with the subject before starting the project; we, as a team, had little experience with 5G. After the midterm, we had thought we finished the research phase and started implementation, but after a few meetings, we discovered new information which required a redesign of our initial protocol presented in the midterm. With the new information, we changed our final goals and pivoted our focus towards

fully describing the protocol and how it could be implemented and away from doing the actual implementation work due to the time constraints of this project.

Final Outcomes

We have completed several diagrams describing our proposed design for building remote attestation into the 5G network in an N2 handover scenario. Our design uses the AKA protocol for attesting the UEs and the CRA protocol for asynchronously attesting the gNBs.

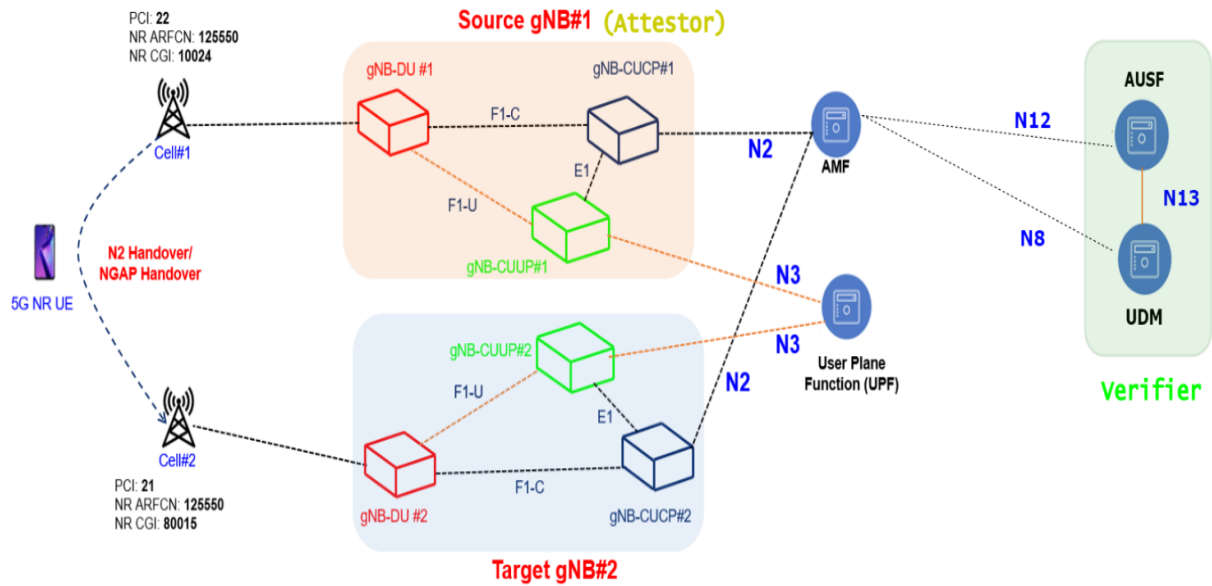


Figure 5: N2 Handover Procedure

Figure 5 shows an overview of the components involved in our N2 handover with mutual attestation protocol. The two components attesting each other are the UE and the target gNB. The AUSF and UDM together serve as the verifier for the attestation.

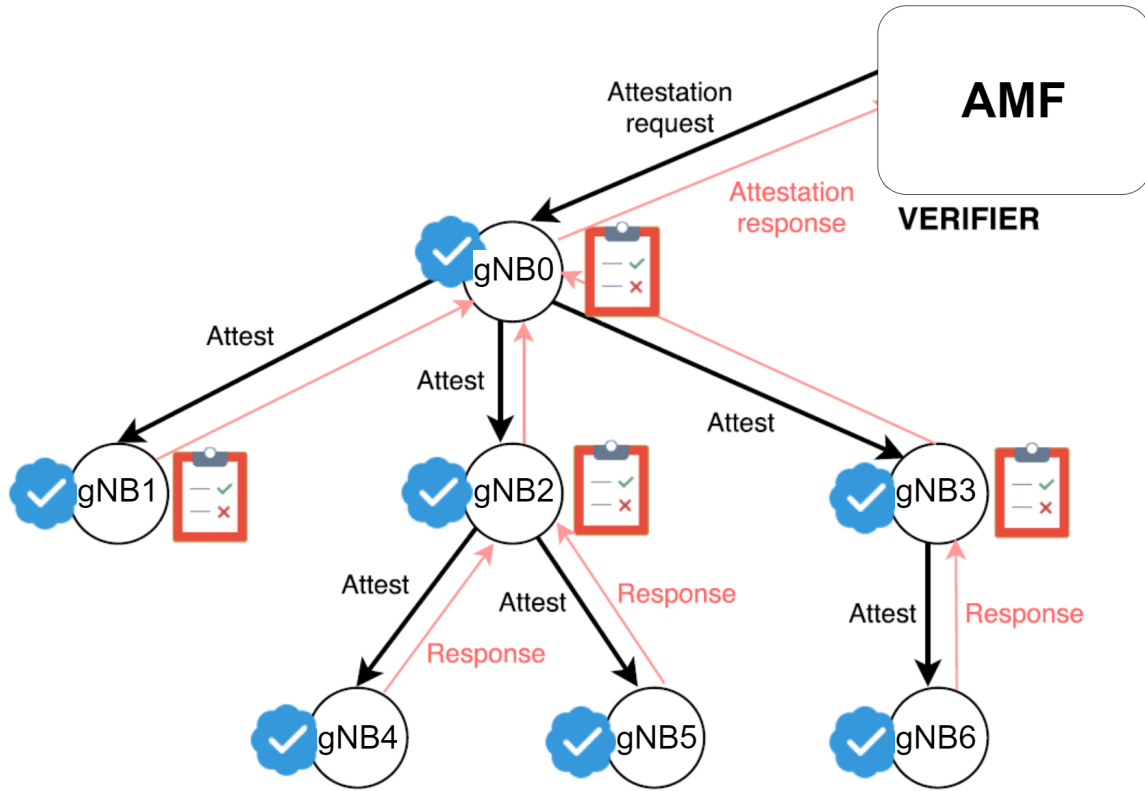


Figure 6: SEDA Protocol

Figure 6 shows the SEDA (Scalable Embedded Device Attestation) protocol, a subsection of the CRA protocol used for scaling remote attestation upwards. SEDA allows the verifier to efficiently perform attestation over a spanning tree, rooted at the verifier. The protocol comprises two phases: an offline phase and an online phase.

The offline phase comprises device initialization and device registration.

Device Initialization is performed by the owner in a trusted setting, which provides the gNB with their expected correct configuration and credentials. Device Registration is where the gNB securely establishes pairwise keys with their neighbors using their credentials.

In the online phase, which is based on creating a spanning tree, the protocol is initialized by the verifier (AMF). The AMF broadcasts an attestation request to the gNB, creating a spanning tree. For each gNB, after verifying the authenticity of the request, the gNB propagates it to other gNB's in the spanning tree. Every gNB then attests the other connected gNB using the shared keys and sends an accumulated result to the AMF indicating whether the subtree has the correct configuration. This process continues along the spanning tree until reaching AMF, which assesses the status of the gNB by verifying only the last attestation report.

SEDA offers some degree of protection against DoS attacks by limiting the “join request” frequency or by making them low-priority tasks in the network. SEDA also proposes an idea

where every device piggybacks the IDs of the compromised devices in the subtree alongside the aggregated attestation result.

SEDA has some important limitations. First, it requires the network to be “stable,” meaning every gNB will maintain a fixed set of neighbors throughout the operation. Second, configurations are deployed statically during the offline phase, and it is unclear how they are updated. Third, the attacker model considered by this protocol is limited to a software-only attacker.

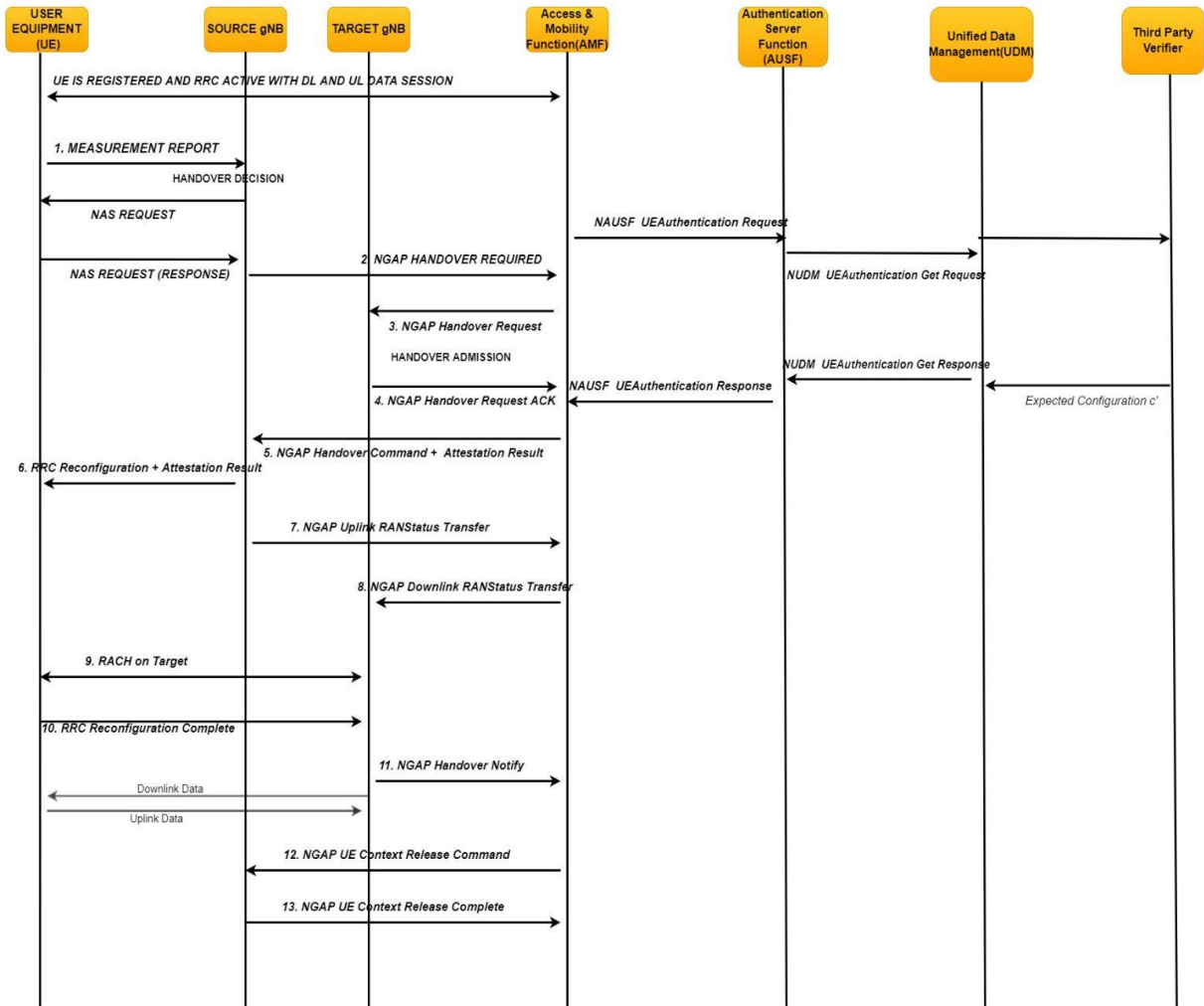


Figure 7: N2 Handover With Remote Attestation

Figure 7 shows the call flow of the N2 handover with the mutual attestation protocol we designed, wherein the source gNB decides the need for handover based on a measurement report sent by the UE. To attest UE, the source gNB requests the UE to provide attestation information along with identity in the form of a NAS request message. The handover-required decision is

then forwarded to AMF. The AMF then checks the availability of target gNB for handover by performing an NGAP handover request-response message. The AMF also relays the attestation information of UE to AUSF embedded in the NAUSF UEAuthentication message. The AUSF then makes a request to the UDM to obtain the expected configuration of the UE device which will be used to attest the UE. Based on the use of a third-party verifier, the UDM receives the expected configuration information for the UE. This information is sent to the AUSF. The AUSF forwards the response to the AMF, which then sends this information as part of the handover command to the source gNB. The source gNB then can proceed with the handover based on the attestation response.

Conclusion

The team initially started to familiarize themselves with 5G and modern attestation protocols by researching topics and getting hands-on experience with 5G. The team then explored and analyzed different attestation workflows and protocols to use in the 5G network.

We decided to use Open5GS to simulate and test the 5G network. We also used Wireshark to collect network data to see the detailed results of the tests we ran. The final product would have been an attestation protocol that worked smoothly and securely in the decentralized 5G network architecture, guaranteeing accurate verification of trust.

However, as the project progressed, the team concluded that developing and describing the complete protocol and implementing the code to test and simulate the protocol would take longer than planned. As a result, we instead focused on completely describing the protocol with diagrams for message call flow and message contents and forgo the implementation for the next team to work on this project.

Future Plans

We have finished the protocol design, so the next step is to integrate and test it over a physical or simulated 5G network. As part of our initial plans, we set up our testbed utilizing various open-source technologies like Docker, Open5GS, and UERANSIM. Although we used Docker for spinning up our testbed, we recommend using Kubernetes because simulating in a production-like environment will make the tests more authentic.

The protocol implementation can be divided into three steps

1. The UE authentication protocol needs to be modified to send the UE's Attestation Value to the 5G Core network where it's managed by the UDM. The network needs to verify the value with a verifier which can be a simple HTTP request.

2. Integrate the CRA protocol to attest the gNBs on the core network.
3. Enable the Mutual Attestation of the UE and target gNB it's connecting to during the N2 Handover procedure. This requires changes in both Open5GS and UERANSIM. In Open5GS, the N2 Handover protocol needs to be modified and as UERANSIM does not yet support N2 Handover, it needs to be implemented prior to making changes for the attestation protocol.

Literature Review

Maat: A Platform Service For Measurement and Attestation [1]

This paper describes Maat as a platform service that provides a centralized policy-driven framework in order to determine the integrity measurement tools and attestation protocols (M&A) required for evaluating the trustworthiness of other software platforms. They achieved features such as authenticity, correctness, protection by involving some of the architectural features such as Support for Multiple Appraisers & attestation protocols, Discrete M&A functions which form the base for its Security Model. In the end, they have shown the applicability and integration of Maat with PAM, NAC, IPsec, and client monitoring.

Orchestrating Layered Attestations [2]

This paper introduces COPLAND as a language for orchestrating layered attestation in response to the need for a flexible language, one that specifies layered attestations, to be used together with a flexible set of attestation mechanisms provided by the Maat framework to measure integrity evidence. COPLAND is designed to meet three requirements: accommodate Maat's wide variety of services, provide formal and unambiguous operational semantics for execution, and determine the trust properties of alternative COPLAND phrases using static analysis through denotational semantics. The COPLAND semantics prove the correctness of attestation outcomes dependent on evidence and the ordering of events that gather evidence. In addition, the COPLAND semantics expand the diversity and complexity of attestation by providing a foundation for testing layered attestation protocols. COPLAND is being tested by Maat and is also being used as the language for remote attestation protocols in other systems.

Flexible Mechanisms for Remote Attestation [3]

This paper covers the many problems that occur when it comes to attestation protocols and different attestation shapes. The main ones it focuses on are issues with the order, evidence types, and needing a flexible model for attestation to be able to interact with different sources that require varying levels of attestation. To this end, it proposes the need for the attestation protocol language COPLAND and provides details of what COPLAND is capable of doing and how it helps resolve such issues.

Practical Runtime Attestation for Tiny IoT Devices [4]

This paper briefly discusses existing attestation protocols and their problems and presents a new attestation protocol primarily focused on low-powered IoT devices. Many attestation protocols exist, some of which require a dedicated hardware module (like a TPM)/Software Mechanism/Both for this process. This paper provides a way to overcome this problem by creating a protocol that utilizes generally available hardware components like an MPU and Flash Memory and makes attestation much more accessible and implementable on existing devices.

A Formal Analysis of 5G Authentication [5]

This paper starts with the problems in existing security protocols and how 5G protocols and tools can help in avoiding this problem. The protocol used is called 5G AKA protocol and the tool used to implement this protocol is called Tamarin Prover.

The protocol is very complex and the tool is used to have a better understanding to obtain a proper output for every single property. The flaws were removed and a faithful model along with security evaluation of 5G AKA was provided.

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Biographical Sketches of Investigators

Rahul Muthyam

Rahul has been involved in the field of computer science for over seven years and is currently pursuing a master's in computer science with a focus in cybersecurity at the University of Texas at Dallas. Rahul is experienced in developing and maintaining software systems using several programming languages like Golang, Java, Javascript, Python, and C and he has also worked extensively with container orchestration platforms like Docker and Kubernetes, which catapulted his knowledge of the inner-workings of computer systems. Rahul is highly interested in cybersecurity, which can be demonstrated by his projects, and he can put his skills and experience to good use and become a valuable team member.

Gandhar Joshi

Gandhar is currently pursuing his master's in computer science with a focus on information assurance at the University of Texas at Dallas. He is familiar with courses such as network security, information security, data, and application security, and various cryptographic protocols. Gandhar's programming skills include Java, SQL, JavaScript, and Linux programming.

Marlon Schumacher

Marlon is currently in his senior year of his bachelor's in computer science and is a fast-track student starting his master's in computer science with a focus in cybersecurity at the University of Texas at Dallas. He is taking courses in computer networks and cyber security. Marlon has experience with Java, C++, C, SQL, JavaScript, PHP, and Python through coursework and internships. He has used GitHub for several group projects and has some exposure to JSON and WireShark.

Benjamin Stark

Benjamin Stark is currently a master's student in computer science with a focus on cybersecurity and information assurance at the University of Texas at Dallas. He has taken courses in networks and network security as well as general information security courses that involve learning and reading protocols. He has also done many projects and competed in challenges involving traffic sniffing; specifically, he has used Wireshark extensively and tcpdump. He has done many projects using C and C++ in his course work and has used GitHub to collaborate on projects with teams.

Nikhil Joshi

Nikhil is currently a master's student in computer science, cybersecurity. This is his first semester here at UTD (The University of Texas of Dallas). Nikhil has taken subjects on network security and system security in his first semester. He is learning new concepts and platforms which will help him in his course and our project as well. Nikhil has done projects in C, C++, Java, and Python in his undergraduate college.